

Introduction to Artificial Intelligence and Machine Learning

School of Engineering
Nanyang Polytechnic

Trainer

Foo Yong Wee, Ph.D.
Manager, School of Engineering
Nanyang Polytechnic

Email: foo_yong_wee@nyp.edu.sg

Introduction to Artificial Intelligence and Machine Learning

Mentimeter

Join at menti.com | use code

8368 7949

Instructions

Go to

www.menti.com

Enter the code

8368 7949



menti.com

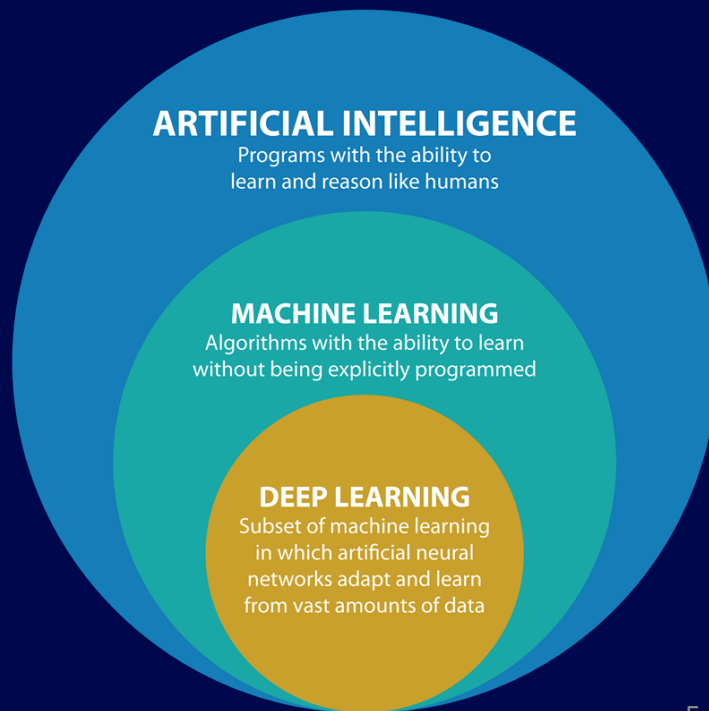
8368 7949

<https://www.mentimeter.com/app/presentation/alp4nm33pue1rtd3mr25qpcoyb68zzr6/present?question=27h6nzjhu4ey>

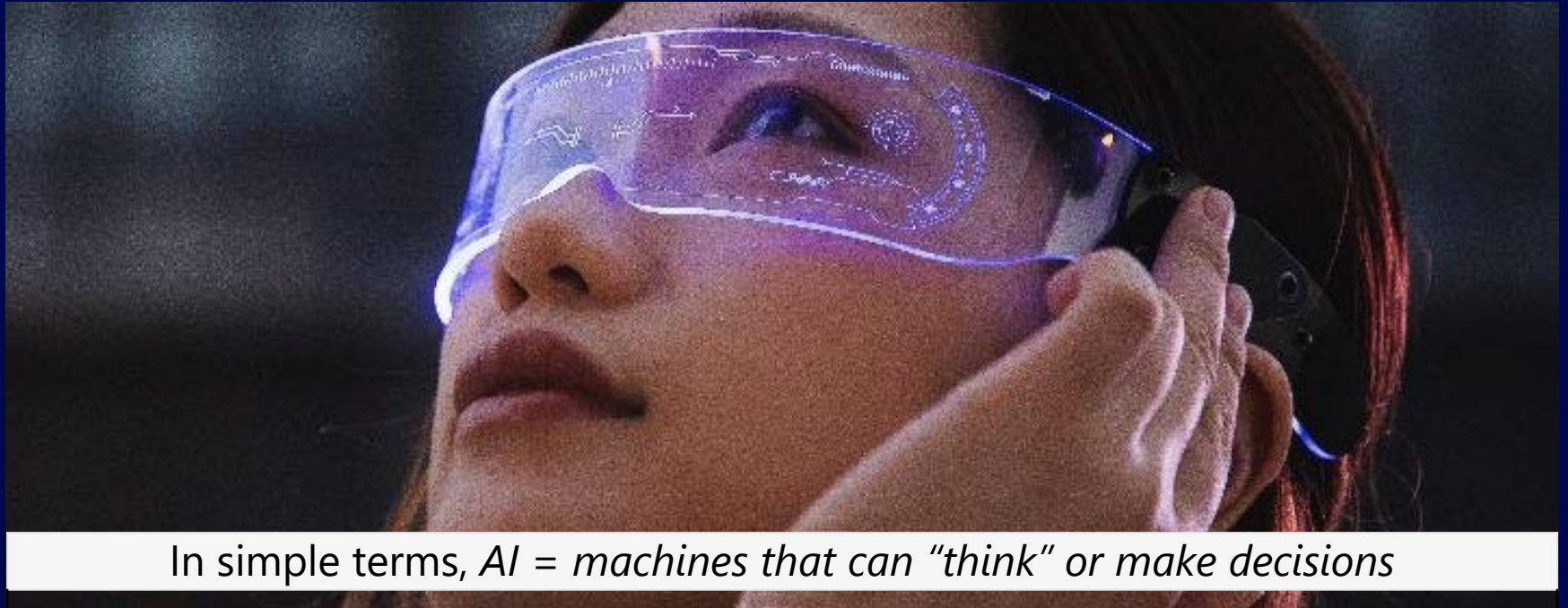
Outline

- 1 What is Artificial Intelligence (AI)
- 2 What is Machine Learning (ML)
- 3 What is Deep Learning (DL)
- 4 Applications of AI and ML

Full material can be found at
https://github.com/yongweefoo/intro_ai_ml



What is AI



In simple terms, *AI = machines that can "think" or make decisions*



Why AI Matters Now

AI is viewed as the “new electricity” – a general-purpose technology. Projected to add nearly **\$20 trillion to global economy by 2030**. It’s driving a new tech boom, with heavy investment from companies to integrate AI for efficiency and new products.

Definition of AI

Artificial Intelligence is the ability of a program or machine to think and behave like a human.

Turing Test

A machine is said to pass the Turing Test when it exhibits behaviour that is indistinguishable from that of a human.

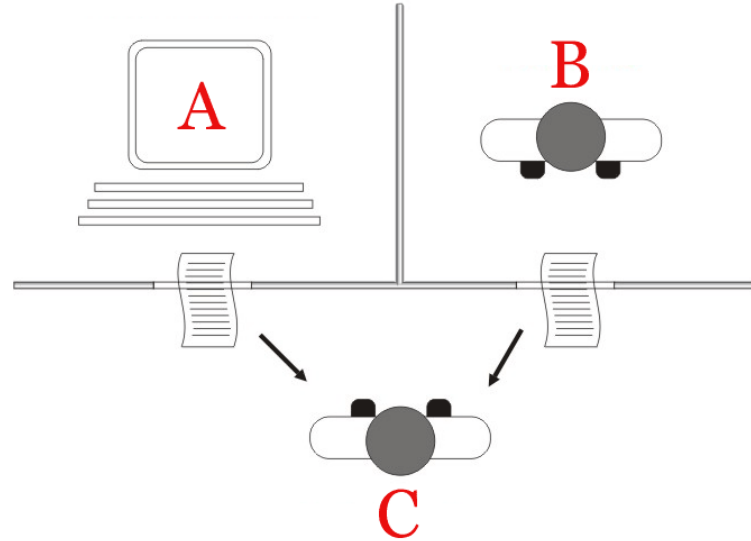


Alan Turing (1912–1954)

The Turing Test



A computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or from a computer



Artificial Intelligence - The **Turing test**, developed by [Alan Turing](#) in 1950₉

History of AI



Microsoft released the first individual intelligent assistant Microsoft Cortana in the world.

Watson won Jeopardy game



Deep Blue beat world chess champion



AlphaGo won Go champion



ChatGPT



Claude

Generative AI



Knowledge engineering
Expert systems
Computer vision
Natural language understanding
Lisp machines
Japan's fifth generation computer project



"... every aspect of learning or any other feature of intelligence can be so precisely described that machine can be made to simulate it."

- John

McCarthy

Workshop on AI at Dartmouth College



The golden years

First AI winter

AI boom

Second AI winter

AI renaissance

AI revolution

1943

1956

1974

1980

1987

1993

1997

2011

2016

2020 2023 2026..

Artificial neurons
(McCulloch and Pitts)

Symbolic processing
Formal representation (logic, ...)
Reasoning, inference
Search and problem solving
Connectionism (neural networks)

... within a generation the problem of creating artificial intelligence will be substantially solved.

- Marvin Minsky



Data mining
knowledge discovery
Machine learning
Cognitive computing
Mathematical/statistical methods
Supercomputers

Big data
Data analytics
Massively distributed computing
Big data distributed databases
Cloud
IOT

AI Resurgence

Computing performance
& Cloud

IOT

Big data

Internet technologies
& Social Media



AI / Machine Learning

Advanced data analytics

Deep learning

Deep neural networks

**BIG DATA, HIGH-PERFORMANCE
COMPUTING AND CLOUD, HAS
TRANSFORMED AI**

AI Boom - Key Players



(founded 2015 by Elon Musk, Sam Altman, etc.)



(founded 2021 by ex-OpenAI
researchers, including Dario Amodei)

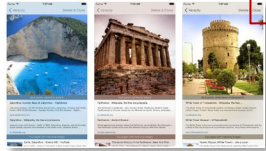


Microsoft 365
Copilot

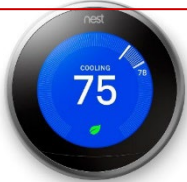


AI is All Around Us

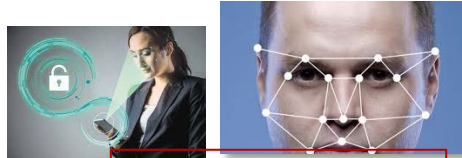
Image search



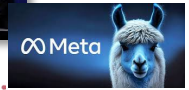
Smart IoT devices



Facial recognition



First chess, then Jeopardy, then Go. Now poker too has fallen to AI



Gemini

ChatGPT



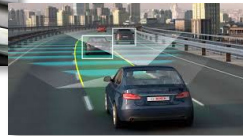
Generative AI



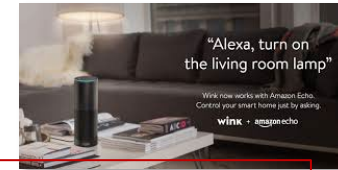
Drones, UAV



Self-driving vehicles



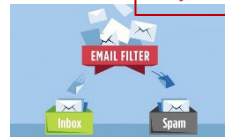
Virtual smart assistants



Recommendation engines



Spam filters



Chatbots

Robo-advisors



Language translation



AI Technologies

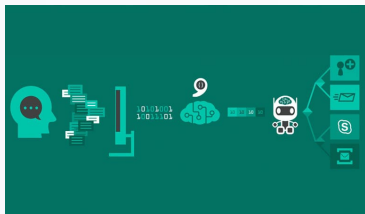
Computer vision: a field of AI that focuses on enabling machines to interpret, analyze and understand visual information such as images and videos.



Speech processing: analyzing, interpreting, and generating human speech using AI techniques including voicing process, statistical features of speech signals, speech recognition, machine-based speech synthesis, and speech perception



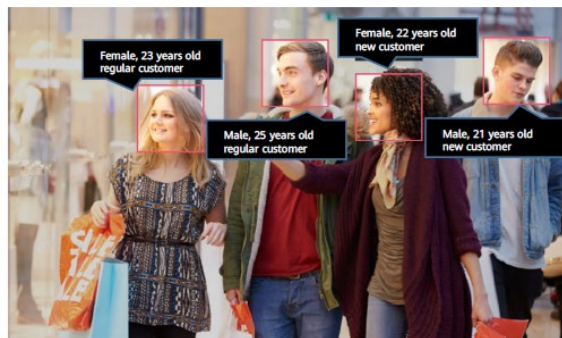
Natural language processing (NLP) and Large Language Model (LLM): the use AI to understand and process text and human language for a range of applications, including language translation, sentiment analysis, chatbots, and content creation.



Generative AI (or Gen AI) generate new, unique data samples that are similar to the examples it was trained on such as image, art, text, music and video generation



Computer Vision



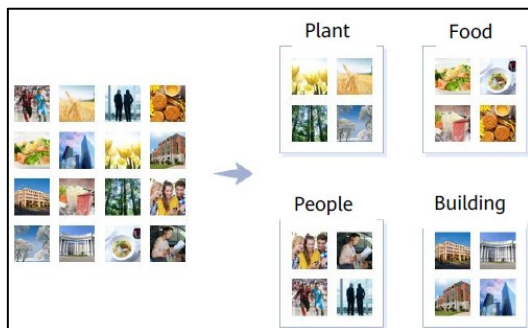
Traffic Analysis

Facial recognition Comparison Gallery Authentication result



Electronic Attendance

Common applications include image classification, target detection, image segmentation, target tracking, optical character recognition (OCR), and facial recognition.



Smart Album

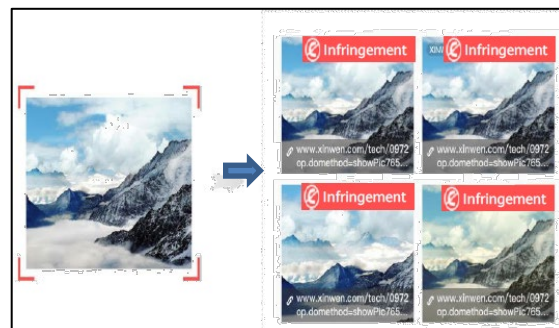


Image Search

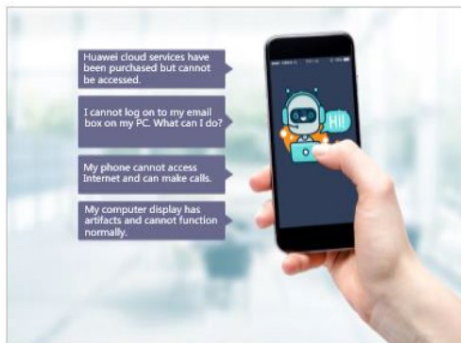


Authentication



Action Analysis

Speech/Audio Processing



Question Answering Bot (QABot)



Voice Navigation



Intelligent Education

Other applications:

- Spoken language evaluation
- Diagnostic robot
- Voiceprint recognition
- Smart sound box
- ...

Common applications include voice recognition, voice synthesis, voice wakeup, voiceprint recognition, and audio-based incident detection.



Real-time Conference Records

Natural Language Processing & Large Language Models



Public Opinion Analysis



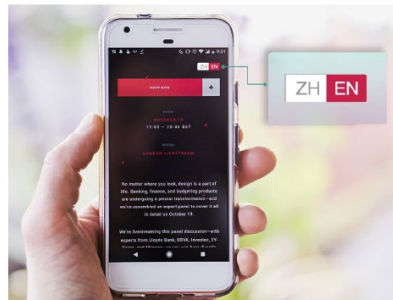
Evaluation Analysis



Text Classification

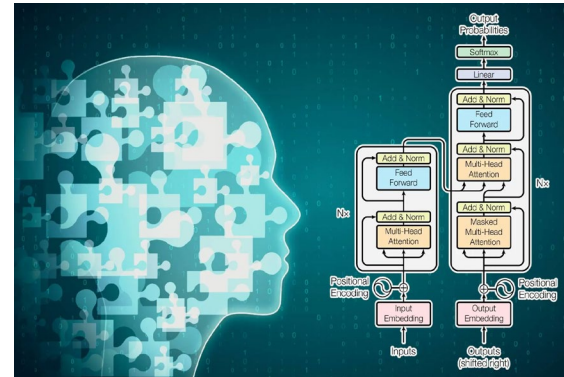


Chatbots

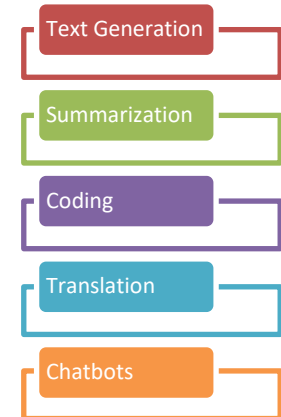


Machine Translation

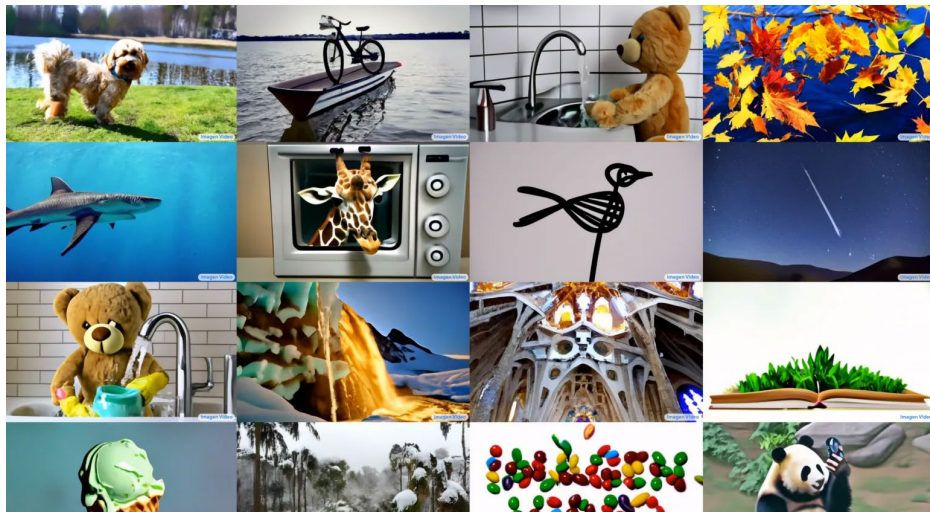
Common applications include machine translation, text mining, content creation, chatbot, text summarization, and sentiment analysis.



Large Language Model, e.g. ChatGPT



Generative AI

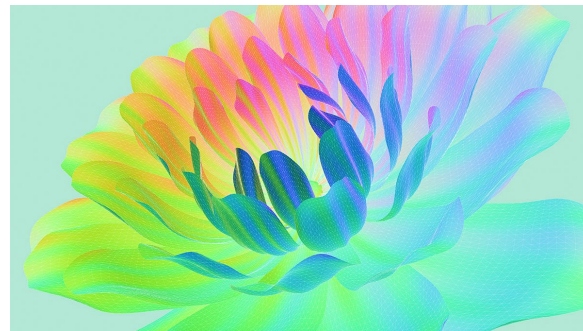


Video Generation

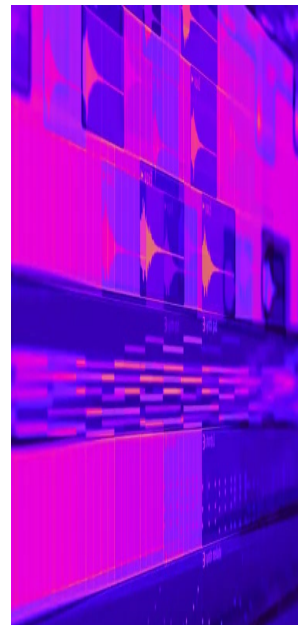
Google's Imagen video (<https://imagen.research.google/video/>) can produce short high-definition video clips using a similar technique to the one it uses to make images. (Photo by Google AI)



Data Augmentation



Art Generation

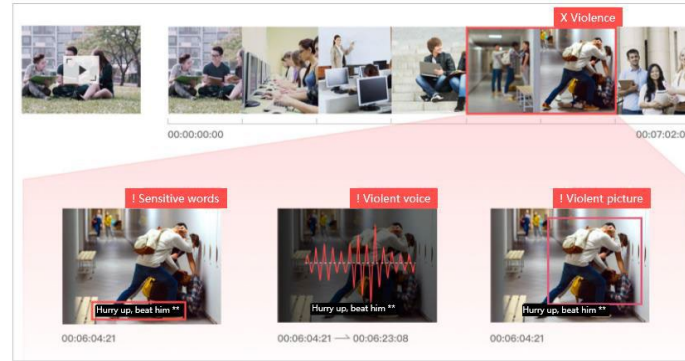


AI-Generated Music

AI Application Fields- Security

Application scenarios:

- **Police use:** suspect identification, vehicle analysis, suspect tracking, suspect search and comparison, and access control at key places
- **Civil use:** facial recognition, warning against potential danger, and home protective measure deployment



Intelligent Security / Security Protection



Face recognition identifying criminals

AI Application Fields - Retail

Unmanned supermarkets of Amazon and Alibaba, use sensors, cameras, computer vision, and deep learning algorithms to completely cancel the checkout process, allowing customers to pick up goods and "just walk out".



Unmanned Store: Amazon Go



Unmanned store : Alibaba

AI Application Fields – Hospitality and Smart Hotel/Home



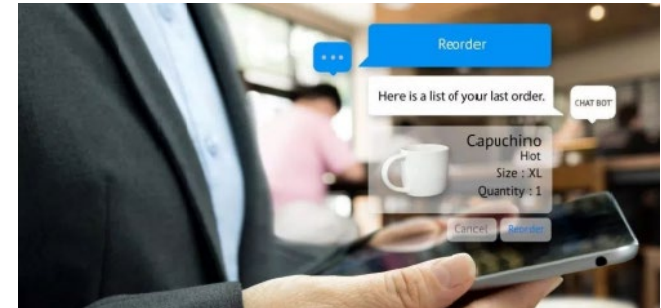
Control smart home products with voice processing such as air conditioning temperature adjustment, curtain switch control, and voice control on the lighting system.

Implement **home security protection** with computer vision technologies, for example, facial or fingerprint recognition for unlocking, real-time intelligent camera monitoring, and illegal intrusion detection.

Develop user profiles and recommend content to users with the help of machine learning and deep learning technologies and based on historical records of smart speakers and smart TVs.

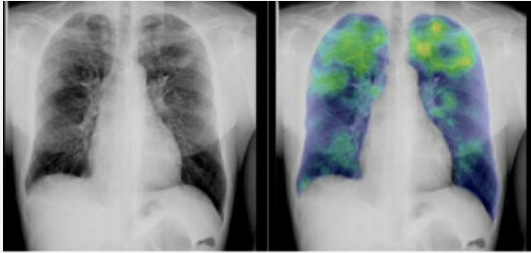


Frontdesk Robots

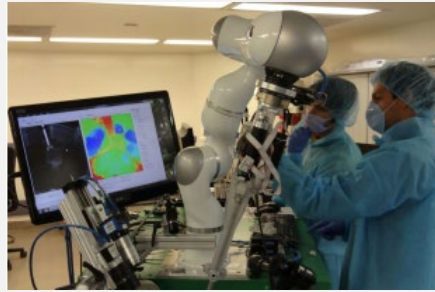


Chatbot for hotel/services

AI Application Fields – Healthcare



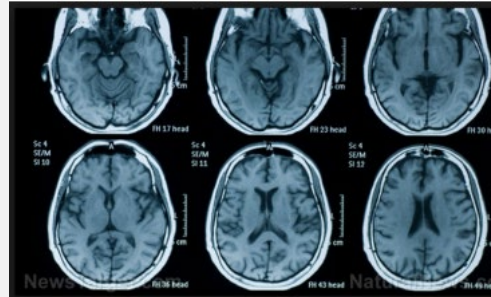
Identifying tuberculosis



Robotics-assisted surgery



Detecting brain bleeds



Detecting Alzheimer's disease

Other applications:

Medicine mining: quick development of personalized medicines by AI assistants

Health management: nutrition, and physical/mental health management

Hospital management: structured services concerning medical records (focus)

Assistance for medical research: assistance for biomedical researchers in research

Virtual assistant: electronic voice medical records, intelligent guidance, intelligent diagnosis, and medicine recommendation

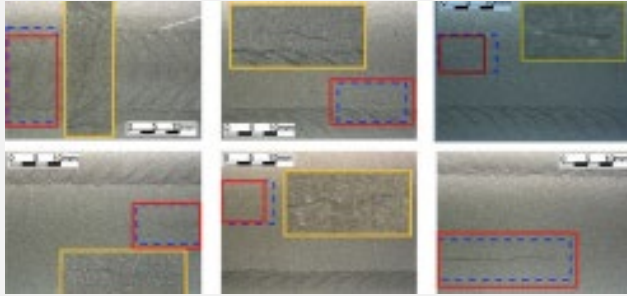
Medical image: medical image recognition, image marking, and 3D image reconstruction

Assistance for diagnosis and treatment: diagnostic robot

Disease risk forecast: disease risk forecast based on gene sequencing



AI Application Fields – Smart Manufacturing



Defect Detection



AI in Pharmaceutical Manufacturing Process



Robots in Mfg



AI Robots in Amazon's Warehouse



AI Application Fields – Auto Industry



Automatic Vehicle Insurance and Loss Assessment

AI technologies help insurance companies optimize vehicle insurance claims and complete vehicle insurance loss assessment using deep learning algorithms such as image recognition.



Autonomous Driving

Currently, only some commercial passenger vehicle models, such as Audi A8, Tesla, and Cadillac, support L2 and L3 Advanced driver-assistance systems (ADAS). L4 and L5 autonomous driving is expected to be first implemented on commercial vehicles in closed campuses. A wider range of passenger vehicles require advanced autonomous driving, which requires further improvement of technologies, policies, and infrastructure. It is estimated that L4 and L5 autonomous driving will be supported by common roads in 2025–2030.

The Society of Automotive Engineers (SAE) in the U.S. defines 6 levels of driving automation ranging from 0 (fully manual) to 5 (fully autonomous). L0 indicates that the driving of a vehicle completely depends on the driver's operation. The system above L3 can implement the driver's hand-off operation in specific cases, L5 depends on the system when vehicles are driving in all scenarios.

Tesla makes push to roll out advanced FSD self-driving in China

BT Published Thu, May 30, 2024 - 06:54 PM

Tesla

Follow



FSD sales in China would open a new source of revenue for Tesla, which has seen its sales volumes there fall by 7.6 per cent in the first four months of the year. PHOTO: REUTERS

TESLA is preparing to register its 'Full Self-Driving' software with authorities in China in the run up to its planned rollout of the technologically advanced feature this year, three people with knowledge of the matter said.

The US electric vehicle maker is also considering selling the software as a monthly subscription to users of its cars in China, its second-largest market, they said.

A successful software registration with China's Ministry of Industry and Information Technology will pave the way for Tesla to internally test Full Self-Driving (FSD) by having its employees drive on China's public roads before delivering it as an upgrade to its Chinese users in the coming months, said two of the people. The company currently offers two, less-advanced versions of its Autopilot driver assistance system in China.

Full-Self Driving (FSD) Car

- Features include:
 - Autopilot
 - Traffic-Aware Cruise Control
 - Autosteer
 - Full Self-Driving (Supervised)
 - Navigate on Autopilot
 - Auto Lane Change
 - Autopark
 - Summon/Smart Summon
 - Autosteer on City Streets
 - Traffic and Stop Sign Control

<https://youtu.be/MA12MNFxwoA?si=-uEI0G3JjKo6BIh0>



Activity – Teachable Machine

Go to the Google Teachable Machine website at <http://teachablemachine.withgoogle.com/> and click on the ‘Get Started’ button. (See diagram).

Teachable Machine

Train a computer to recognize your own images, sounds, & poses.

A fast, easy way to create machine learning models for your sites, apps, and more – no expertise or coding required.

Get Started

New Project

 Open an existing project from Drive.

 Open an existing project from a file.

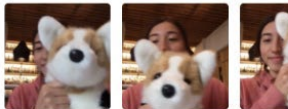
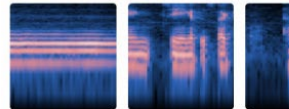


Image Project

Teach based on images, from files or your webcam.



Audio Project

Teach based on one-second-long sounds, from files or your microphone.



Pose Project

Teach based on images, from files or your webcam.

Quiz

What is the Turing test?

- A. A test to determine whether a machine can exhibit human-like intelligence
- B. A test to determine whether a machine can understand natural language
- C. A test to determine whether a machine can learn from data
- D. A test to determine whether a machine can generate data

Quiz

Which of the following is NOT an example of AI?

- A. Use of computer vision to do away with the grocery checkout process
- B. Sorting department store items by bar code
- C. Identifying tuberculosis in an x-ray image based
- D. Recommend movies based on users' watching history

Quiz

Which of the following is NOT an application example of Large Language Models (LLMs)?

- A. Perform as a chatbot
- B. Generate codes
- C. Controlling audio system volume
- D. Summarize text

Discussion Question

- How could artificial intelligence transform your industry, and what are some concrete examples of its potential applications and impacts?

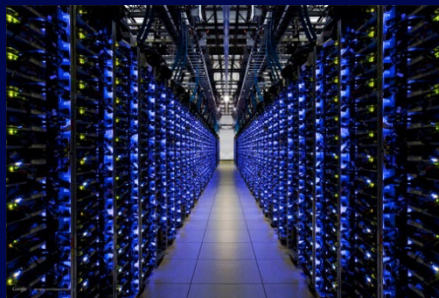
What is Large Language Model (LLM)

- An AI model that is trained on large amounts of text to identify patterns between words, concepts, and phrases so that it can generate responses to prompts.

Training



Chunk of the internet ~ 10TB of text



6,000 GPUs for 12 days, ~1e24 FLOPS



140 GB

*numbers for Llama 2 70B

What is LLM (Cont'd)



140 GB

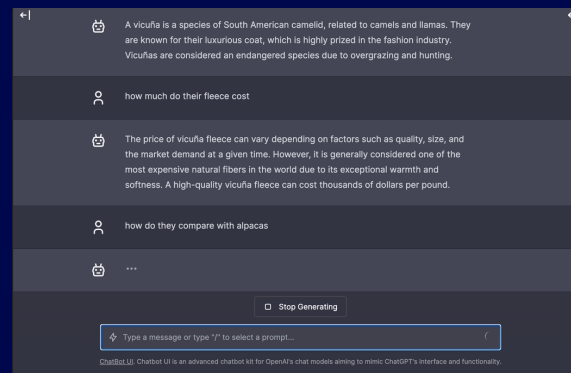


run .c
~ 500 lines

Llama-2 (70B)



Inference



Core LLM Technology

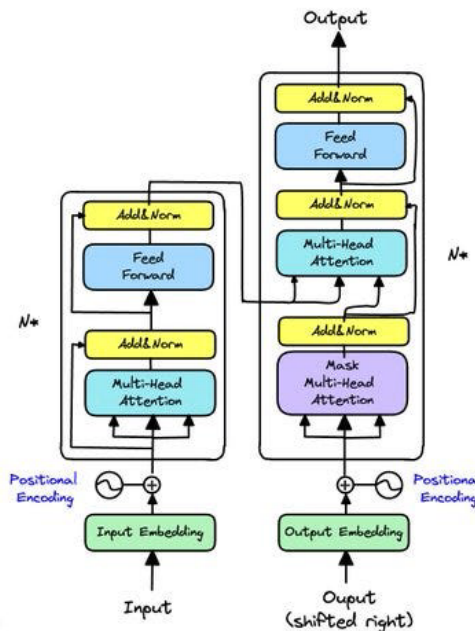
Attention Is All You Need

Ashish Vaswani* Google Brain avaswani@google.com	Noam Shazeer* Google Brain noam@google.com	Niki Parmar* Google Research nikip@google.com	Jakob Uszkoreit* Google Research uszko@google.com
Llion Jones* Google Research llion@google.com	Aidan N. Gomez[†] University of Toronto aidan@cs.toronto.edu	Lukasz Kaiser* Google Brain lukaszkaiser@google.com	
Illa Polosukhin[‡] illia.polosukhin@gmail.com			

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

*Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representation and became the other person involved in nearly every detail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Lukasz and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research.
[†]Work performed while at Google Brain.
[‡]Work performed while at Google Research.



“The cat that the dog chased was brown”

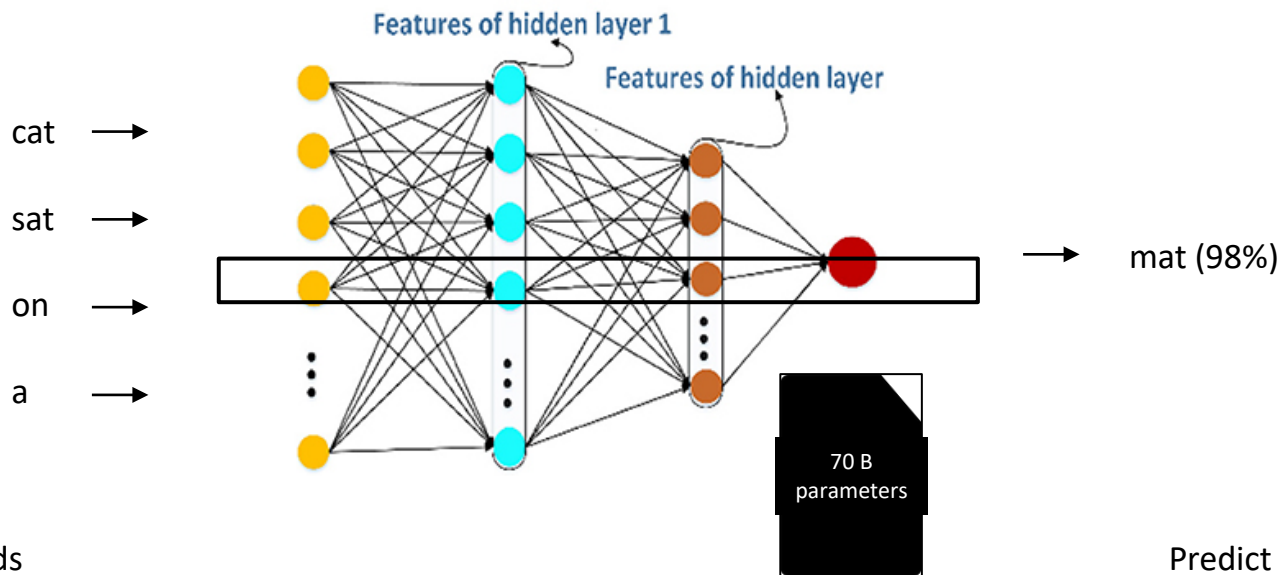
Embeddings & Vector Databases – it’s about similarity and relationships

Attention (The Magic Behind GPT) – it’s about understanding context

Transformers Model

How does LLM work

Neural Network Transformer (simplified for illustration)



The network adjust the billions of parameters iteratively adjusting them to make better prediction (measure)

How does LLM work

- Identifies patterns in human languages
- Predict the word most likely to come next in a sequence of words (complete the sentence)
- Uses statistics to analyse the relationships between all the words in a given sequence
- Computes the probability for thousands of possible words to come next in that sequence
- The predictive power enables LLMs to response to question/requests whether to complete a simple sentence or to develop a compelling business strategy, or a persuasive new product launch or a creative ad campaign

Examples of LLMs

- Popular examples are ChatGPT, Google Gemini, Microsoft CoPilot, Meta LLAMA, Anthropic Claude, and more. For example:
 - ChatGPT at <https://chatgpt.com/>
 - Claude at <https://claude.ai/>
 - Copilot at <https://copilot.microsoft.com/>
 - Gemini at <https://gemini.google.com/>
 - Llama at <https://llama.meta.com/>

What is Prompt Engineering

- Is the practice of developing effective prompts that elicit useful output from generative AI
- **Key rule:** The clearer and more specific the prompts, the more likely you are to get useful output
- **Helpful tip:** Use iteration to evaluate the output and revising your prompts
- **Techniques:** Use techniques such as few-shot prompting
- **Required:** Critical thinking and creativity (a skill that can be practiced)

LLM Limitations

- Although LLMs are powerful, you may not always get the output you want
- It may generate information that is skewed. For instance, the LLM is trained on bias data (e.g. websites, articles reflects unfair biases)
- Or it might not generate sufficient content about a specific domain or topic because the data it was trained on does not contain enough information about that topic
- Or it has the tendency to hallucinate – AI outputs that are not true. This means LLM can generate texts that are factually inaccurate
- The same prompt may not generate the same output in another instance
- Therefore, LLM, while a power AI tool, requires human guidance for effective use

How to write prompts

- **Key rule:** Quality of the prompt affects the quality of the output
- **Prompt design:** The design the prompt to get the best output you want
- Prompt examples:
“Generate a list of 5 potential themes for an event” vs
“Generate a list of 5 potential themes for a professional conference on customer experience in the hospitality industry.”
- The latter prompt has specific, relevant context and instructions
- *Note: There will be instance where you will not get a quality output regardless of the quality of your prompt due to the limitations of LLMs*

Prompt Examples: Information Retrieval

- Question Answering: "What is the capital of France?"
- Summarization: "Summarize the key findings of this research paper."
- Topic Exploration: "Provide five interesting facts about space exploration."

Prompt Examples: Creative Writing/Content Creation

- Story Starter: "Once upon a time, in a world where..."
- Poem Generator: "Compose a poem in the style of [famous poet] about [theme or topic]."
- Script Writing: "Create a movie scene showcasing a tense conversation between two characters."

Prompt Examples: Code Generation

- Function Creation: "Write a Python function that takes two numbers as input and returns their sum."
- Code Translation: "Translate this Java code into Python."
- Bug Fixing: "Fix the syntax error in this code snippet."

Prompt Examples: Translation

- Language Translation: "Translate this sentence from English to Spanish."
- Dialect Conversion: "Rewrite this text in British English."
- Formal/Informal Conversion: "Write a more formal version of this email."

Activity – Prompt Engineering

Text generation prompts (poem, story, email)

1) Poem

Write a 12-line poem about learning AI as an adult.

Tone: hopeful, lightly humorous.

Include: one vivid metaphor, one Singapore reference, and an uplifting ending.

2) Short story

Write a 100-word story for adult learners new to AI.

Theme: “From fear to curiosity.”

Structure: 2 scenes (before / turning point / after).

Style: simple language, vivid details, no jargon.

End with a one-sentence lesson.

3) Appeal email

Draft a polite appeal email to request approval for a 2-hour AI literacy workshop for staff.

Audience: department head.

Include: purpose, benefits, proposed date/time, minimal resources needed, and a clear call to action.

Tone: respectful, confident, concise (≤ 180 words).

Image generation prompt

1) Minimal-token image prompt

Photorealistic image: adult learner in a modern classroom, laptop open, friendly AI hologram on screen.

Warm lighting, shallow depth of field, candid moment, 35mm lens, high detail.

2) Style + constraints

Flat vector poster, “AI for Everyone”.

Simple shapes, 3-color palette (teal, white, coral), clean typography, inclusive diverse adults, minimal background, high contrast.

Activity – Prompt Engineering (Cont'd)

Correct / improve a paragraph

1) Improve writing + explain changes

Improve the paragraph below for clarity and professionalism.

Then:

- list 5 specific edits you made (bullet points)
- explain 2 improvements in tone

Keep meaning unchanged.

Paragraph: [PASTE TEXT]

Assess a short English essay using a prescribed rubric

You are an English writing assessor. Grade the essay using this rubric:

Rubric (0–5 each):

- 1) Task fulfillment
- 2) Organization & coherence
- 3) Vocabulary
- 4) Grammar & accuracy
- 5) Tone & audience awareness

Output format:

- Scores (0–5) with 1–2 sentence justification each
- 3 strengths (bullets)
- 3 improvement areas (bullets)
- A revised thesis statement (1 sentence)
- 2 targeted practice exercises

Essay: [PASTE ESSAY]

Sample paragraph (for correction / improvement prompts)

AI is becoming more important in many jobs today. However, many adults feel unsure or even scared when they hear about AI. They worry that AI is too technical or that it may replace their jobs. With proper guidance and practical examples, AI can instead become a helpful tool that supports daily work and improves productivity.

Sample short essay (for assessment with rubric)

Essay title: *Learning AI as an Adult*

Learning artificial intelligence as an adult can be challenging but rewarding. Many adults return to learning after many years, and they may feel anxious about new technology. AI often seems complex because of unfamiliar terms and rapid changes.

However, when AI is introduced in a simple and practical way, it becomes easier to understand. Adults usually bring strong work experience and problem-solving skills, which help them see how AI can be applied in real situations. For example, AI tools can help with writing, summarising information, and generating ideas more efficiently.

In conclusion, adults should not avoid learning AI because of fear. With the right support and mindset, learning AI can build confidence and open new opportunities in the workplace.

AI Assistants vs AI Agents



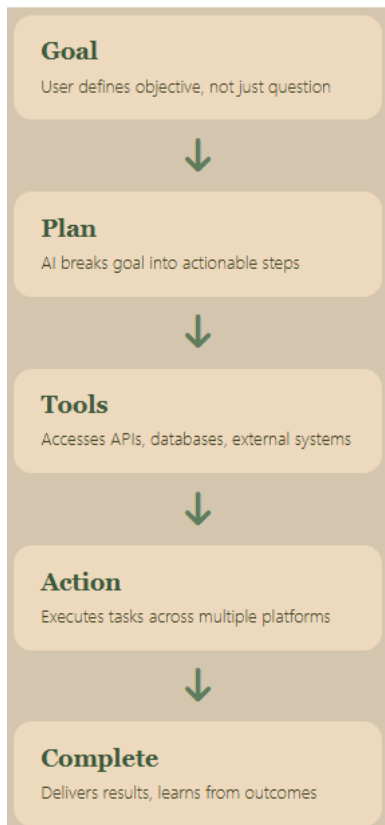
They can take initiative, pursue goals over time, and perform sequences of actions by itself.

They typically wait for user prompts and then respond.



- Agentic AI operates autonomously through continuous reasoning loops that plan, execute, and adapt until goals are achieved
- Agents think in ReAct loops that continuously reason, act, observe results, and adjust strategy
- Unlike chatbots, agents proactively identify needs and take autonomous actions across integrated systems
- Modern agents possess short-term task context and long-term memory enabling continuous learning
- Can orchestrate complex workflows by calling APIs, querying data warehouses, and coordinating across systems

Agentic AI





Ethical AI to Mitigate its Potential Risks

- Developing robust AI governance frameworks and regulations
- Promoting transparency and explainability in AI systems
- Conducting thorough ethical reviews of AI applications
- Investing in AI safety research
- Educating the public about AI capabilities, limitations, and its potential abuse (e.g. scam, manipulation)

End of Chapter 1